SELECT Statements:

```
mydb=# SELECT * FROM employees;
 emp_id | first_name | last_name |       position       |  s
alary  | hire_date  | dept_id
--------+------------+-----------+----------------------+---
-------+------------+---------
      1 | John       | Smith     | Software Engineer    | 75
000.00 | 2026-10-12 |         1
      2 | Jane       | Doe       | Senior Developer     | 85
000.00 | 2026-10-12 |         1
      3 | Mike       | Brown     | DevOps Engineer      | 80
000.00 | 2026-10-12 |         1
      4 | Sarah      | Davis     | Marketing Specialist | 55
000.00 | 2026-10-12 |         2
(4 rows)

mydb=# SELECT first_name, position FROM employees;
 first_name |       position
------------+----------------------
 John       | Software Engineer
 Jane       | Senior Developer
 Mike       | DevOps Engineer
 Sarah      | Marketing Specialist
(4 rows)

mydb=# SELECT first_name, salary, salary * 0.10 AS potential
_bonus FROM employees;
 first_name |  salary  | potential_bonus
------------+----------+-----------------
 John       | 75000.00 |       7500.0000
 Jane       | 85000.00 |       8500.0000
 Mike       | 80000.00 |       8000.0000
 Sarah      | 55000.00 |       5500.0000
(4 rows)
```

WHERE Clause:

```
mydb=# SELECT e.first_name, e.salary
mydb-# FROM employees e JOIN departments d ON e.dept_id = d.
dept_id
mydb-# WHERE d.dept_name = 'Marketing';
 first_name |  salary
------------+----------
 Sarah      | 55000.00
(1 row)

mydb=#
mydb=# SELECT e.first_name
mydb-# FROM employees e JOIN departments d ON e.dept_id = d.
dept_id
mydb-# WHERE e.salary > 75000 AND d.dept_name = 'Engineering
';
 first_name
------------
 Jane
 Mike
(2 rows)

mydb=#
mydb=# SELECT product_name, price FROM products WHERE price
BETWEEN 20 AND 50;
 product_name | price
--------------+-------
 Strawberry   | 25.00
 Blueberry    | 45.50
 Headphones   | 35.00
(3 rows)
```

PATTERN Matching (LIKE, ILIKE):

```
mydb=# SELECT customer_name FROM customers WHERE customer_na
me LIKE 'Bra%';
 customer_name
---------------
 Brad
 Brandon
(2 rows)

mydb=# SELECT product_name FROM products WHERE product_name
LIKE '%berry';
 product_name
--------------
 Strawberry
 Blueberry
(2 rows)
```

```
mydb=# SELECT email FROM users WHERE email ILIKE '%@gmail.co
m';
        email
----------------------
 gh45278@gmail.com
 ghdh4893@gmail.com
 usg@gmail.com
 aidana.k@gmail.com
 bekzat.t@gmail.com
 nurzat88@gmail.com
 aibek.n@gmail.com
 dinara.zh@gmail.com
 kanat.b@gmail.com
 meerim.a@gmail.com
 sultan.k@gmail.com
 timur99@gmail.com
 marat.b@gmail.com
 anna.smith@gmail.com
 ruslan.t@gmail.com
 danial.z@gmail.com
 madina.a@gmail.com
 john@gmail.com
 MARY@Gmail.com
(19 rows)
```

REGULAR Expressions:

```
mydb=# SELECT first_name FROM employees WHERE first_name ~ '
^[JM]';
 first_name
------------
 John
 Jane
 Mike
(3 rows)

mydb=# SELECT email FROM users WHERE email ~* '^[A-Z0-9._%+-
]+@[A-Z0-9.-]+\.[A-Z]{2,}$';
         email
-----------------------
 gh45278@gmail.com
 ghdh4893@gmail.com
 usg@gmail.com
 aidana.k@gmail.com
 bekzat.t@gmail.com
 nurzat88@gmail.com
 aibek.n@gmail.com
 dinara.zh@gmail.com
 kanat.b@gmail.com
 meerim.a@gmail.com
 sultan.k@gmail.com
 elena.v@mail.ru
 timur99@gmail.com
 aigerim.m@outlook.com
 marat.b@gmail.com
 anna.smith@gmail.com
 ruslan.t@gmail.com
 klara.k@yandex.ru
 danial.z@gmail.com
 madina.a@gmail.com
 john@gmail.com
 MARY@Gmail.com
(22 rows)
```

IN and EXISTS Operators:

```
mydb=# SELECT first_name, dept_id FROM employees WHERE dept_
id IN (1, 2);
 first_name | dept_id
------------+---------
 John       |       1
 Jane       |       1
 Mike       |       1
 Sarah      |       2
(4 rows)

mydb=#
mydb=# SELECT first_name FROM employees
mydb-# WHERE emp_id IN (SELECT employee_id FROM sales WHERE
amount > 1000);
 first_name
------------
 John
 Mike
 Jane
(3 rows)

mydb=#
mydb=# SELECT first_name FROM employees e
mydb-# WHERE EXISTS (
mydb(#   SELECT 1 FROM sales s
mydb(#   WHERE s.employee_id = e.emp_id AND s.amount > 1000
mydb(# );
 first_name
------------
 John
 Mike
 Jane
(3 rows)
```

CASE Statements:

```
mydb=# SELECT first_name, salary,
mydb-# CASE
mydb-#   WHEN salary > 80000 THEN 'Senior'
mydb-#   WHEN salary BETWEEN 60000 AND 80000 THEN 'Mid-Level
'
mydb-#   ELSE 'Junior'
mydb-# END AS employee_level
mydb-# FROM employees;
 first_name |  salary  | employee_level
------------+----------+----------------
 John       | 75000.00 | Mid-Level
 Jane       | 85000.00 | Senior
 Mike       | 80000.00 | Mid-Level
 Sarah      | 55000.00 | Junior
(4 rows)
```

Common Table Expression (CTEs):

```
mydb=# WITH department_averages AS (
mydb(#   SELECT dept_id, AVG(salary) AS avg_salary
mydb(#   FROM employees
mydb(#   GROUP BY dept_id
mydb(# )
mydb-# SELECT e.first_name, e.salary, e.dept_id, da.avg_sala
ry
mydb-# FROM employees e
mydb-# JOIN department_averages da ON e.dept_id = da.dept_id

mydb-# WHERE e.salary > da.avg_salary;
 first_name |  salary  | dept_id |     avg_salary
------------+----------+---------+--------------------
 Jane       | 85000.00 |       1 | 80000.000000000000
(1 row)
```